

National Science & Technology Examination 2026

Standard Multiple Choice Assessment — Total Questions: 30 | Time: 45 Minutes

Section A: Computer Science & Architecture

1. Which data structure operates on a First-In-First-Out (FIFO) basis?
 - A. Stack
 - B. Queue
 - C. Binary Tree
 - D. Hash Table
2. In computer architecture, what does ALU stand for?
 - A. Arithmetic Logic Unit
 - B. Advanced Link Unit
 - C. Array Logic Universal
 - D. Application Level Utility
3. Which memory management scheme allows non-contiguous physical address space allocation?
 - A. Contiguous Allocation
 - B. Paging
 - C. Swapping
 - D. Fixed Partitioning
4. What is the standard default port used for HTTP communication?
 - A. 21
 - B. 80
 - C. 443
 - D. 8080
5. Which SQL clause is used to retrieve unique records from a database table?
 - A. DISTINCT
 - B. UNIQUE
 - C. GROUP BY
 - D. DIFFERENT
6. In Python, which of the following built-in collection types is immutable?
 - A. List
 - B. Dictionary
 - C. Tuple
 - D. Set
7. Which cryptographic algorithm is an example of asymmetric key cryptography?
 - A. AES
 - B. RSA
 - C. DES
 - D. Blowfish

8. What is the average time complexity of searching an element using Binary Search?

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n \log n)$

9. In web development, what does CSS stand for?

- A. Cascading Style Sheets
- B. Creative System Sheets
- C. Computer Styled Syntax
- D. Central Styling Server

10. Which type of RAM requires periodic electrical refreshing to retain stored data?

- A. Dynamic RAM (DRAM)
- B. Static RAM (SRAM)
- C. ROM
- D. Flash EEPROM

Section B: Software Engineering & Systems

11. What condition is essential in a recursive function to prevent an infinite call stack?
 - A. Loop counter
 - B. Base case
 - C. Global variable
 - D. Exception handler
12. Which Unix/Linux terminal command is used to display the current working directory path?
 - A. pwd
 - B. cwd
 - C. dir
 - D. path
13. In Git version control, which command saves staged changes into the local repository history?
 - A. git push
 - B. git stage
 - C. git commit
 - D. git record
14. What is the standard default port number used by the HTTPS protocol?
 - A. 22
 - B. 53
 - C. 80
 - D. 443
15. Which protocol resolves human-readable domain names into numerical IP addresses?
 - A. DHCP
 - B. DNS
 - C. ARP
 - D. ICMP
16. Which machine learning paradigm learns patterns strictly from labeled training examples?
 - A. Supervised Learning
 - B. Unsupervised Learning
 - C. Reinforcement Learning
 - D. Self-Supervised Learning
17. What cloud service delivery model provides virtualized hardware, storage, and networking on demand?
 - A. SaaS
 - B. IaaS
 - C. PaaS
 - D. FaaS
18. Which logic gate produces an output of 0 only when all its inputs are 1?
 - A. AND
 - B. OR
 - C. NAND
 - D. XOR

19. What is the exact value of 2 raised to the power of 10 (2^{10})?

- A. 512
- B. 1000
- C. 1024
- D. 2048

20. What is the chemical formula for water?

- A. CO₂
- B. H₂O
- C. NaCl
- D. CH₄

Section C: General Science & Modern Technologies

21. What is the approximate speed of light in a vacuum?
- A. 150,000 km/s
 - B. 300,000 km/s
 - C. 450,000 km/s
 - D. 600,000 km/s
22. Which greenhouse gas is released in significant quantities through human fossil fuel combustion?
- A. Argon
 - B. Helium
 - C. Carbon Dioxide
 - D. Neon
23. Which phase of the Software Development Life Cycle (SDLC) directly follows Requirement Analysis?
- A. Deployment
 - B. System Design
 - C. Maintenance
 - D. User Testing
24. In data analysis with Python, which library provides high-performance DataFrame structures?
- A. Pandas
 - B. Matplotlib
 - C. Scikit-learn
 - D. PyTorch
25. Which software tool is primarily used for creating and running lightweight containerized applications?
- A. VirtualBox
 - B. Docker
 - C. VMware ESXi
 - D. Wine
26. In RESTful API design, which HTTP method is considered both idempotent and safe for reading resources?
- A. POST
 - B. PATCH
 - C. GET
 - D. DELETE
27. What typically causes a Stack Overflow runtime error in executing programs?
- A. Deep or infinite recursion
 - B. Heap memory fragmentation
 - C. Null pointer dereference
 - D. Disk space exhaustion
28. What is the first compilation phase that breaks source code into tokens?
- A. Syntax Analysis
 - B. Lexical Analysis
 - C. Semantic Analysis
 - D. Code Optimization

29. In modern microprocessors, what unit is most commonly used to measure CPU clock frequency?
- A. Megabytes (MB)
 - B. Gigahertz (GHz)
 - C. Terabytes (TB)
 - D. Milliseconds (ms)
30. What type of cyber attack tricks individuals into disclosing confidential credentials via fraudulent emails?
- A. Phishing
 - B. SQL Injection
 - C. Denial of Service (DoS)
 - D. Buffer Overflow

Official Answer Key

Section-by-section correct option reference for automated scoring validation.

Questions 1 - 10	Questions 11 - 20	Questions 21 - 30
1. B	11. B	21. B
2. A	12. A	22. C
3. B	13. C	23. B
4. B	14. D	24. A
5. A	15. B	25. B
6. C	16. A	26. C
7. B	17. B	27. A
8. B	18. C	28. B
9. A	19. C	29. B
10. A	20. B	30. A